

Linear Algebra II

FMA_2F001_EP

Lecture Notes

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1 Class 1: Basic Arithmetic

The arithmetic fact $1 + 1 = 2$ was shown by Russel and Whitehead. The proof operates using Peano's axioms for the natural numbers $\mathbb{N}$.

Theorem 1: Peano's Axioms for $\mathbb{N}$

- **Axiom 1:** $0 \in \mathbb{N}$.
- **Axiom 2:** There exists a successor function $S : \mathbb{N} \rightarrow \mathbb{N}$.

From these, we can define $1 = S(0)$ and $2 = S(S(0))$.

Definition 2: Addition

Addition is defined by the following rules:

1. $\forall m \in \mathbb{N} : m + 0 = m$ (Add1).
2. $\forall m, n \in \mathbb{N} : m + S(n) = S(m + n)$ (Add2).

Example 3: Computing $1 + 1$

$$1 + 1 = 1 + S(0) \stackrel{(\text{Add2})}{=} S(1 + 0) \stackrel{(\text{Add1})}{=} S(1) = 2.$$

Theorem 4: Peano's Axiom 9 (Induction)

Let P be a unary predicate. To prove $\forall m \in \mathbb{N}, P(m)$ is true, it is enough to prove:

- $P(0)$ is true.
- $\forall m : P(m) \implies P(S(m))$.

Theorem 5: Well Ordering Principle (WOP)

Every non-empty subset of $\mathbb{N}$ has a minimum.

Proposition 6: WOP implies Induction

The Well Ordering Principle can be used to prove the principle of Induction.

PROOF. Let $S = \{m \mid P(m) \text{ is not true}\}$. If $P(m)$ is not true for every m , then S is non-empty. We apply WOP to S , meaning there is a minimum element $m_0 = \min S$. But $m_0 \neq 0$, so $m_0 = S(m)$ for some m . This creates a contradiction. $\square$

1.1 Relations on a Set

Definition 7: Relations and Equivalence

Let S be a set with elements $x, y \in S$. A relation on S is defined as a subset of $S \times S$. An equivalence relation must satisfy all three of the following properties:

- **Reflexivity:** $\forall x \in S, x \sim x$.
- **Symmetry:** $\forall x, y \in S, x \sim y \iff y \sim x$.
- **Transitivity:** $\forall x, y, z \in S, x \sim y \text{ and } y \sim z \implies x \sim z$.

Example 8: Equivalence Relations

“Same Parity” and “Same remainder mod n ” are equivalence relations.

Remark 9: Partitions

An equivalence relation is the same as a partition of S .

Definition 10: Modular Classes

For every $n \in \mathbb{N}$, $a \equiv b \pmod{n}$ defines an equivalence relation. The set of equivalence classes for this relation is denoted $\mathbb{Z}/n\mathbb{Z}$.

1.2 Modular Arithmetic and Divisibility

Theorem 11: Chinese Remainder Theorem (Sunzi)

If m and n are coprime, there is a bijection between $\mathbb{Z}/mn\mathbb{Z}$ and $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$.

Example 12: Applying the Chinese Remainder Theorem

For $n = 2$ and $m = 3$, we can map between $\mathbb{Z}/6\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. If we want to find all $m \in \mathbb{Z}$ such that $m \equiv 2 \pmod{5}$ and $m \equiv 3 \pmod{7}$, the solution is the set $\{14 + 35k \mid k \in \mathbb{Z}\}$.

Theorem 13: Euclidean Division

$\forall a, \forall b \neq 0, \exists (q, r)$ such that $a = bq + r$ and $0 \leq r < |b|$.

Theorem 14: Bézout's Theorem

$\exists (u, v) \in \mathbb{Z}^2$ such that $au + bv = \gcd(a, b)$.

Example 15: Euclid's Algorithm

Applying the algorithm to find coefficients for 5 and 7:

- $7 = 1 \times 5 + 2.$
- $5 = 2 \times 2 + \gcd(5, 7).$
- $1 = 5 - 2 \times 2 = 5 - 2 \times (7 - 5) = 3 \times 5 - 2 \times 7.$

We have found that $3 \times 5 \rightarrow (0, 1)$ and $-2 \times 7 \rightarrow (1, 0)$. If we want to find m such that $m \equiv 2 \pmod{5}$ and $m \equiv 3 \pmod{7}$, we compute $(3 \times 5) \times 3 + (-2 \times 7) \times 2$.

1.3 Prime Numbers

Definition 16: Prime Number

A prime number is a number $p \in \mathbb{N}$ such that $p \geq 2$ and p has only divisors 1 and p .

Theorem 17: Infinitude of Primes

There are infinitely many primes.

Theorem 18: Tao's Theorem

At least one of the following sets of primes (p, q) is infinite:

- **Twin primes:** $|p - q| = 2.$
- **Cousin primes:** $|p - q| = 4.$
- **Sexy primes:** $|p - q| = 6.$

Remark 19: Goldbach Conjecture

The Goldbach conjecture posits that every even number is equal to $p + q$ with p and q primes.